Lab Experiment -

Implement your own ping program.

AIM:

To implement a simple ping program using Python

Algorithm:-

i) Import required modules and define ping function.

ii) check os and select -n for windows or -c for Linux.

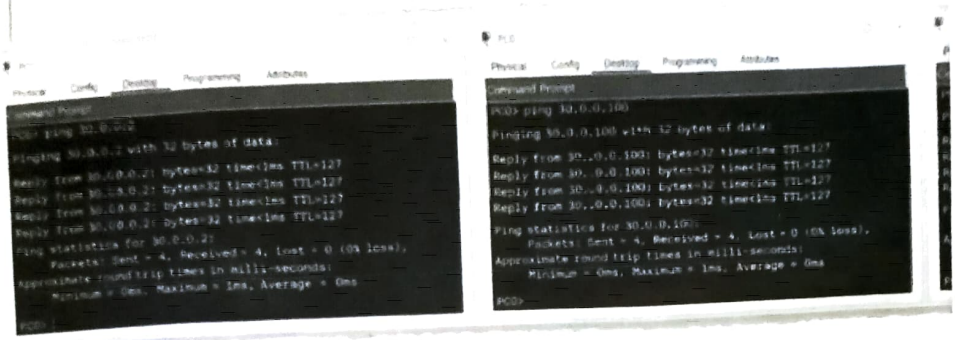
iii) create and execute ping and using subprocess

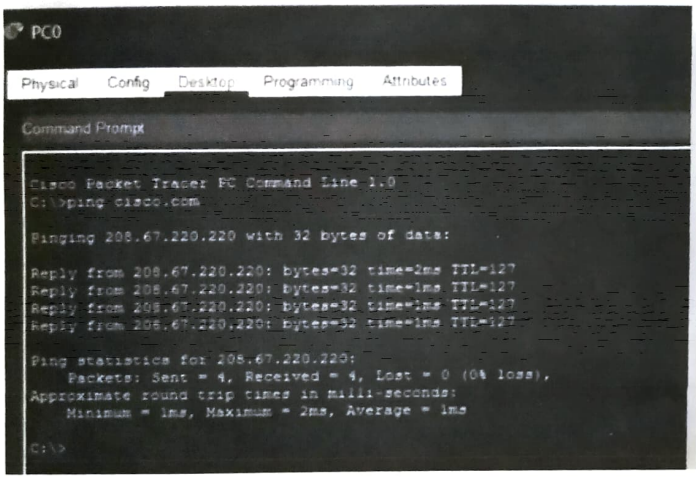
iv) Display ping result or error message.

Program:-

```
import os
import platform
import subprocess
def ping(host = "google.com", count=4):
    if platform.system().lower()=="windows":
        param = "-n"
    else:
        param = "-c"
    command = ["ping", param, str(count), host]
try:
    output = subprocess.check_output(
        cmd,
```

```
universal_newlines=True)
print(output)
except exception as e:
    print("Ping failed:", e)
if __name__ == "__main__":
    ping("8.8.8.8")
```





32, time = 20ms

=32 time 21ms TTL=

Result -

Thus, Ping program was implemented successfully

Lab Experiment

Raw socket packet sniffing

AIM:

To write Python program using RAW sockets to capture and display network packets.

Algorithm:

i) Create raw socket using socket module
ii) Bind socket to network interface
iii) receive packets continously using recv fromc) fn.
iv) Display captured packet info.

Program:

```
import socket
s = socket.socket(
        socket.AF_INET,
        socket.SOCK_RAW,
        socket.IPPROTO_IP
    )
host = socket.gethostbyname
    (socket.gethostname())
s.bind((host, 0))
s.setsockopt(socket.IPPROTO_IP, socket.IP_
                HPRINCL, 1)
Print("Packet sniffer started...")
while true:
```

```
Packet, address = s.recvfrom(65535)
Print("Packet received from: ", address)
Print("Packet data: ", packet)
```

I/P:
No direct input is required. program captures packets from network.

O/P:
Packet sniffer started...

Result:
Thus, packet sniffing program using RAW sockets was successfully implemented.

Lab experiment

Web log analysis using webalizer

Aim:

To analyze different types of web logs using webalizer tool

Procedure:

i) Run windows version of webalizer
ii) Select/input web log file downloaded from web
iii) click run webalizer
iv) view generated monthly statistics & info.

Output:

Monthly statistic
Host

13 1/2 10%

Result:

Thus, web logs were successfully implem